that another realization of $\mathbb{C}^3_-$ is by action of S_4 by symmetries (not necessarily rotations) of the regular tetrahedron. Yet another realization of this representation is the space of functions on the set of four elements (on which S_4 acts by permutations) with zero sum of values.

4.4. Duals and tensor products of representations

If V is a representation of a group G, then V^* is also a representation, via

$$\rho_{V^*}(g) = (\rho_V(g)^*)^{-1} = (\rho_V(g)^{-1})^* = \rho_V(g^{-1})^*.$$

The character is $\chi_{V^*}(g) = \chi_V(g^{-1})$.

We have $\chi_V(g) = \sum \lambda_i$, where the λ_i are the eigenvalues of g in V. If G is finite, these eigenvalues must be roots of unity because $\rho(g)^{|G|} = \rho(g^{|G|}) = \rho(e) = \mathrm{Id}$. Thus for complex representations

$$\chi_{V^*}(g) = \chi_V(g^{-1}) = \sum \lambda_i^{-1} = \sum \overline{\lambda_i} = \overline{\sum \lambda_i} = \overline{\chi_V(g)}.$$

In particular, $V \cong V^*$ as *representations* (not just as vector spaces) if and only if $\chi_V(g) \in \mathbb{R}$ for all $g \in G$.

If V, W are representations of G, then $V \otimes W$ is also a representation, via

$$\rho_{V\otimes W}(g) = \rho_V(g) \otimes \rho_W(g).$$

Therefore, $\chi_{V\otimes W}(g) = \chi_V(g)\chi_W(g)$.

An interesting problem discussed below is decomposing $V \otimes W$ (for irreducible V, W) into the direct sum of irreducible representations.

4.5. Orthogonality of characters

We define a positive definite Hermitian inner product on $\mathrm{F}_c(G, \mathbb{C})$ (the space of central functions) by

$$(f_1, f_2) = \frac{1}{|G|} \sum_{g\in G} f_1(g)\overline{f_2(g)}.$$